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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Multi-Disciplinary Courses (MDC) - MG2MDCBTG100 - TOOLS ANDTECHNIQUES IN BIOTECHNOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 Hours **Maximum Marks: 35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I)and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A-One Word Answer

Answer any 10 questions-Each question carries 1 marks

1. Which enzyme prevents DNA supercoiling? [U] / [CO1]

   Topoisomerase/Gyrase-1 mark
2. How do restriction exonucleases differ from restriction endonucleases? [K] / [CO2]

   Restriction Exonucleases- 0.5 marks, restriction endonucleases- 0. 5 marks
3. Define real time PCR. [K] / [CO1, CO4]

   Definition- 1 mark
4. Define Electrophoresis. [K] / [CO1]

   Definition- 1 mark
5. Name two methods of gene cloning. [U] / [CO2]

   Any two methods- 1 mark
6. Comment on the purpose of NCBI. [U] / [CO3]

   NCBI-expansion- .5 marks, purpose- .5 marks.
7. Cite the methods of site directed mutagenesis. [U] / [CO2]

   Any two methods- 1 mark.
8. Describe annealing in PCR. [U] / [CO1, CO4]

   Step (temperature)- 1 mark
9. What are the steps involved in expressing a protein in a host cell by gene cloning? [U] / [CO2].

   Gene cloning- steps- 1 mark.
10. Mention the primary focus of bioinformatics? [K] / [CO3]

    Any one-1 mark.
11. What factors influence the migration rate of proteins in SDS-PAGE? [K] / [CO1]

    Any two factors-1 mark.
12. Identify the anionic stain used in SDS-PAGE. [K] / [CO4].

    Coomaassie brilliant blue- 1 mark.

Part B-Short Essay

Answer any 3 questions-Each question carries 5 marks

13. Discuss the limitations of both AGE and PAGE in electrophoretic separation. [K] / [CO1]
    AGE—2.5 marks, PAGE- 2.5 marks
14. Outline the main stages in Central Dogma? [K] / [CO1]
    Central Dogma- definition- 1 mark, stages- explanation- 4 marks.
15. What is the role of t-RNA in gene expression? [U] / [CO2].
    t-RNA- introduction- 1 mark, Role in gene expression- 4 marks.
16. Describe the process of translation from initiation to termination. [U] / [CO1]
    Translation- introduction- 1 mark, process- steps- explain- 4 marks.
17. Examine the principle of mass spectrometry. [K] / [CO4]
    Mass spectrometry-definition-2 marks, principle – 3 marks

Part C-Essay

Answer any 1 questions-Each question carries 10 marks

18. Describe the organization of DNA. [K] / [CO1].
    DNA- introduction- 2 marks, Organization- 6 marks, diagram- 2 marks.
19. Describe the differences between bacteriophage vectors and cosmid vectors. What are their advantages and limitations?
20. Cloning vectors- introduction-2 marks, features – 3 marks, bacteriophage vectors and cosmid vectors- explain- 5 marks [U] / [CO2]